



# Welcome to FIL Engineering

Graduate Training Program for New Joiners — Product Engineering, Cyber Security & AI Foundations

Product Engineering

CI/CD & Platform

Cyber Security

AI Engineering

FIL Ecosystem

Facilitated by Abinash Kumar Mishra · Co-Founder, AceRanker AI · Chief AI Consultant , TekMentors Consulting

# Product Engineering Training Program

*End-to-end software engineering excellence — nine modules across a common foundation, a specialisation track and the FIL ecosystem*

## Common Foundation

Modules 1 – 4

Product mindset · Core engineering practices · CI tools & workflows · Cyber security basics for all

## Specialisation Track

Modules 5 – 6 or 7 – 8

AI Engineering track | Cyber Security track

## FIL Ecosystem

Module 9

AWS / Azure · Terraform · Kubernetes · CD pipeline · Front end & databases



### Program Structure

- Duration: 8 weeks — common base, then AI or Security specialisation
- Format: morning theory, afternoon hands-on labs & pair programming
- Languages: Java & Python throughout
- Approach: one banking product built, secured and shipped end-to-end
- Platform: FIL-standard tooling — GitHub, Jenkins, Terraform, AWS, Kubernetes



### Core Philosophy

- Think like a product engineer — business outcomes and value stream first
- Quality-first: clean, testable, modular code from day one
- Security embedded in every SDLC stage, not bolted on
- AI as an everyday engineering tool — and a system to be engineered responsibly
- Speed and agility through DevOps practices that scale



### Learning Outcomes

- Ship changes through FIL CI/CD — Jenkins shared libraries, GitHub Action workflows, ArgoCD
- Provision governed infrastructure with Terraform inner-source modules
- Build, evaluate and operate an LLM / RAG service (AI track)
- Detect, triage and respond to security incidents (Security track)
- Run a service on AWS & Kubernetes with observability and audit hooks

# Common Foundation — Modules 1 to 4

*Every new joiner completes these four modules before specialising*



## MODULE 1

### Product Engineering Mindset

- What it means to be a product engineer
- Quality-first thinking
- Business outcomes & value-stream focus
- User experience
- Speed and agility
- DevOps practices to scale



## MODULE 2

### Core Engineering Practices

- Why good engineering practices matter
- Code & repository standards (incl. FIL standards)
- Clean, testable, modular code · Test-driven development
- Continuous Integration
- Emergent & evolutionary design · Refactoring
- Pair programming
- Inner source engineering at FIL — Terraform inner-source modules, developer portal & catalogs



## MODULE 3

### CI Tools & Workflows

- Continuous Deployment
- GitHub Actions & Jenkins — building and configuring pipelines
- Shared libraries & workflows: FIL Jenkins Shared Libraries, FIL GitHub Action Workflows
- Terraform basics — infrastructure provisioning
- API design & versioning
- Observability tooling



## MODULE 4

### Cyber Security Basics for All

- Fundamentals — CIA triad, common threats, attack lifecycle
- Securing the SDLC: design → code → test → deploy → operate
- Secure coding, dependencies, secrets management
- Prompt injection, data-leakage model, API access control
- Web app security — OWASP Top 10, SQL injection & XSS demos
- DevSecOps — SAST / SCA / secret-scanning gates, dependency updates
- Cloud security & identity — IAM, MFA, Zero Trust, Azure / AWS basics

# Specialisation Tracks — Modules 5 to 8

*After the common foundation, each graduate follows one track*

## AI ENGINEERING TRACK



### MODULE 5

#### AI Foundation for Engineers

- AI / LLM fundamentals — models, training vs inference, tokens, embeddings, context window
- Calling AI APIs — hands-on in Java & Python; responses, streaming, error handling
- Prompt engineering — crafting techniques, structured inputs / outputs, cost & latency
- AI-assisted development — GitHub Copilot hands-on, AI agents in DevOps workflows
- Responsible AI · Fidelity AI Agents deep dive · live pre-configured use-case walkthrough



### MODULE 6

#### Applied AI Engineering

- ML & LLM foundations — ML lifecycle, datasets / features, train-validation-test splits, LLM app architecture
- RAG & vector databases lab — embeddings in practice, building a RAG pipeline from a starter repo
- Evaluation & model selection — quality metrics, hallucination checks, cost / performance, fallbacks
- Deployment patterns — API design for AI services, authentication, rate limits, caching
- MLOps / LLMOps — versioning, experiment tracking, prompt & config management, drift & quality monitoring, incident handling



### MODULE 7

#### Network, Governance & Compliance

- Networking basics — TCP/IP, DNS, HTTP/S, ports, firewalls
- Identity & access — MFA, SSO, least privilege, privileged accounts
- Security policies, asset inventory, data classification, acceptable use
- OS security — Linux & Windows users, permissions, processes
- Governance, risk & compliance — GDPR / PDPA, PCI-DSS; security assurance & FinTech risk



### MODULE 8

#### Defensive Engineering, Detection & Response

- Vulnerability scanning & penetration testing
- API security testing; secrets & credential exposure
- Container / orchestration attack surface; cloud misconfigurations
- SIEM fundamentals — ingestion, normalisation, correlation, alerting
- Logging, alerting & triage; email and web security

# Module 9 — FIL Ecosystem: AWS / Azure, Kubernetes & CD Pipeline

*How code actually ships at FIL — the platform every graduate lands on*



## Terraform & Governance

- Terraform deep dive · Infrastructure as Code
- Account bootstrapping & governance
- Inner-source modules deep dive



## FIL on AWS

- IAM · Networking with Amazon VPC
- Containerisation on AWS — FIL patterns for hosting APIs for internal & external consumption
- Monitoring, logging & troubleshooting



## Kubernetes Deep Dive

- K8s concepts & manifests
- Helm · Kustomize · ArgoCD
- Secrets management



## CI & CD for K8s Applications

- Integration mechanisms with Jenkins, ArgoCD & GitHub Actions
- Traceability & audit hooks



## Front End

- JavaScript & TypeScript essentials for full-stack delivery



## Databases

- Databases overview
- Usage for unit testing & API testing

Toolchain across the program

Java

Python

JUnit

Pytest

GitHub Actions

Jenkins

Terraform

Docker

Kubernetes

Helm

ArgoCD

AWS

Azure

Copilot

Vector DB

SIEM